

CHAPTER 1 PROOF

1.1 Proof by contradiction

- **Definition / Core Concept**

- **Contradiction:** A disagreement between two statements, signifying that both cannot be simultaneously true.
- **Negation (Notation):** A statement that asserts the falsehood of another statement.

- **Rational Number ($\mathbb{Q}$):** A number that can be written in the form $\frac{a}{b}$, where a and b are integers.

- **Irrational Number:** A number that cannot be expressed in the form $\frac{a}{b}$, where a and b are integers.

- **Governing Principle: The Method of Proof by Contradiction**

- To prove a statement is true, start by assuming it is **not true** (assume its negation).
- Use logical steps to show this assumption leads to something **impossible**.
- This impossibility must be either:
 - A contradiction of the initial assumption.
 - A contradiction of a known fact.
- **Conclusion:** The initial assumption was incorrect, therefore the original statement must be **true**.

- **Conceptual and Mechanistic Explanation (Logic Chains)**

- **Logic of Parity:**
 - Even numbers are multiples of 2; form: $2k$.
 - Odd numbers are one more than a multiple of 2; form: $2k + 1$.
 - $\text{odd} + \text{even} = \text{odd}$.
- **Logic of Primality:**
 - Assume a finite list of all primes: $p_1, p_2, p_3, \dots, p_n$.
 - Construct $N = (p_1 \times p_2 \times p_3 \times \dots \times p_n) + 1$.
 - Dividing N by any prime in the list yields a remainder of 1.
 - Contradiction: N must be prime or have a prime factor not in the "complete" list.

- **Assumptions, Conditions, and Constraints**

- **Simplest Form Constraint:** In proofs involving irrationality (e.g., $\sqrt{2}$), it is assumed the fraction $\frac{a}{b}$ cannot be reduced further, meaning a and b have **no common factors**.
 - **Integer Requirement:** Variables a , b , n , and k used in defining parity or rationality must be integers.
 - **Equations and Derivations**
 - **Theorem: If n^2 is even, then n is even.**
 - Assumption (Negation): There exists a number n such that n^2 is even but n is odd.
 - Step 1: Write odd n as $n = 2k + 1$.
 - Step 2: $n^2 = (2k + 1)^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$.
 - Result: n^2 is odd, contradicting the assumption that n^2 is even.
 - **Theorem: $\sqrt{2}$ is irrational.**
 - Assumption: $\sqrt{2} = \frac{a}{b}$ (where $a, b \in \mathbb{Z}$ and have no common factors).
 - Step 1: $2 = \frac{a^2}{b^2} \implies a^2 = 2b^2 \implies a$ is even.
 - Step 2: Let $a = 2n$. Then $(2n)^2 = 2b^2 \implies 4n^2 = 2b^2 \implies 2n^2 = b^2 \implies b$ is even.
 - Result: a and b share common factor 2, contradicting the assumption of no common factors.
 - **Practical Significance**
 - **Cryptography:** Proof by contradiction and the properties of very large prime numbers are used to encode chip and pin transactions.
 - **Diagram Integration**
 - [Diagram: Logical flow chart of proof by contradiction: Start -> Assume Negation -> Logical Steps -> Reach Impossibility -> Conclude Original is True]
 - [Diagram: Grid representation of vector-style logic for odd/even parity proofs]
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **General Notations and Sets:**
 - Rational Number Set = $\mathbb{Q}$
 - Rational Form = $\frac{a}{b}$ where $a, b \in \mathbb{Z}$
 - Irrational Number $\neq \frac{a}{b}$

- **Parity Definitions:**
 - Even Number = $2k$ where $k \in \mathbb{Z}$
 - Odd Number = $2k + 1$ where $k \in \mathbb{Z}$
 - **Specific Proof Constants and Expressions:**
 - $n + 2 > n$ (Used in "no greatest odd integer" proof)
 - $(2k + 1)^2 = 4k^2 + 4k + 1$
 - $\sqrt{2}$
 - $a^2 = 2b^2$
 - $N = p_1 \times p_2 \times p_3 \times \dots \times p_n + 1$
 - $\sqrt{3}$
 - $\sqrt{2}$
 - $21a + 14b = 1$
 - $3x + 18y = 1$
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ULTRA-CONDENSED FINAL SUMMARY

- **Proof by Contradiction Strategy:** To prove P , assume not P . Derive a logical impossibility (contradiction). Conclude P is true.
- **Negation Concept:** The starting point of the proof is the exact logical opposite of the target statement.
- **Key Definitions for Proofs:**
 - **Rational:** Ratio of two integers (a/b).
 - **Irrational:** Cannot be a ratio of two integers.
 - **Even:** $2k$; **Odd:** $2k + 1$.
- **Standard Contradiction Results:**
 - $\sqrt{2}$ is irrational (contradicts "simplest form" assumption).
 - Primes are infinite (contradicts "complete list" assumption).
 - No greatest odd integer (always possible to find $n + 2$).
- **Error Check:** When analyzing student proofs, ensure the negation is correctly stated and the contradiction is logically derived from that negation.

CHAPTER 2 PARTIAL FRACTIONS

2.1 Partial fractions

- **Definition / Core Concept**

- **Partial Fractions:** Splitting a single fraction with a compound denominator into the sum of two or more simpler fractions.
- **Constants:** Unknown values (e.g., A , B , C) that must be determined to complete the identity.
- **Governing Law / Principle**
 - A single fraction with two **distinct linear factors** in the denominator can be split into two separate fractions with linear denominators.
 - **General Form:**
$$\frac{5}{(x+1)(x-4)} \equiv \frac{A}{x+1} + \frac{B}{x-4}.$$
- **Conceptual and Mechanistic Explanation (Methods)**
 - **Substitution:**
 - Set the original fraction identical to the sum of partial fractions.
 - Multiply through by the denominator to equate the numerators.
 - Substitute specific values of x that make one of the linear factors zero to eliminate one constant and solve for the other.
 - **Equating Coefficients:**
 - Expand the numerator on the right-hand side after finding a common denominator.
 - Collect like terms (x terms and constant terms).
 - Create simultaneous equations by equating the coefficients of x and the constant terms from both sides of the identity.
- **Assumptions, Conditions, and Constraints**
 - The denominator must contain **distinct linear factors** for this basic method.
 - **Watch out:** This method cannot be used directly for a repeated linear factor in the denominator.
- **Special Cases / Extensions**
 - **More than two distinct linear factors:** The method extends to three or more factors.
 - **Example Form:**
$$\frac{7}{(x-2)(x+6)(x+3)} \equiv \frac{A}{x-2} + \frac{B}{x+6} + \frac{C}{x+3}.$$
- **Equations and Derivations**
 - **Step-by-step logic (Substitution):**
 4. Let $\frac{6x-2}{(x-3)(x+1)} \equiv \frac{A}{x-3} + \frac{B}{x+1}$
 5. $6x-2 \equiv A(x+1) + B(x-3)$
 6. Let $x=3 \implies 16=4A \implies A=4$ (eliminates B)
 7. Let $x=-1 \implies -8=-4B \implies B=2$ (eliminates A).
- **Practical Significance**
 - Partial fractions are used for **binomial expansions** and **integration**.
 - Used to analyze **linear differential systems**, such as **resonant circuits** in electrical systems.

2.2 Repeated factors

- **Definition / Core Concept**
 - **Repeated Linear Factor:** A linear factor in the denominator that is raised to a power (e.g., $(x + 3)^2$).
- **Governing Law / Principle**
 - A single fraction with a repeated linear factor in the denominator can be split into two or more separate fractions where the repeated factor appears with increasing powers in the denominators.
- **Conceptual and Mechanistic Explanation**
 - If the denominator contains $(x + 3)^2$, the partial fractions must include terms for both $(x + 3)$ and $(x + 3)^2$.
 - **General Form:**
$$\frac{2x + 9}{(x - 5)(x + 3)^2} \equiv \frac{A}{x - 5} + \frac{B}{x + 3} + \frac{C}{(x + 3)^2}.$$
- **Equations and Derivations**
 - **Equating Numerators:**
$$11x^2 + 14x + 5 \equiv A(x + 1)(2x + 1) + B(2x + 1) + C(x + 1)^2.$$
 - **Finding Constants:**
 - Substitute root values (e.g., $x = -1$) to find one constant.
 - Equate coefficients of the highest power (e.g., x^2 terms) to find remaining constants if substitution does not cover all of them.
- **Diagram Integration**
 - [Diagram: Fraction with arrow pointing to repeated factor $(x + 3)^2$ showing it splitting into two partial fractions with denominators $(x + 3)$ and $(x + 3)^2$].

2.3 Improper fractions

- **Definition / Core Concept**
 - **Improper Algebraic Fraction:** A fraction where the degree (highest power) of the numerator is **greater than or equal to** the degree of the denominator.
 - **Top-heavy:** Term used for fractions where the degree of the numerator $\geq$ degree of the denominator.
- **Governing Law / Principle**
 - An improper fraction must be converted into a **mixed fraction** (a polynomial plus partial fractions) before further manipulation.
- **Conceptual and Mechanistic Explanation (Conversion Methods)**
 - **Algebraic Division:** Divide the numerator by the denominator.
 - **The Relationship:** $F(x) = Q(x) \times \text{divisor} + \text{remainder}.$
- **Special Cases / Behavioral Conditions**

- **Equal Degree:** If numerator and denominator are both quadratics, dividing produces a constant: $A + \frac{B}{x+1} + \frac{C}{x-1}$.
 - **Numerator Degree > Denominator Degree:** If a cubic is divided by a quadratic, it produces a linear function: $Ax + B + \frac{C}{x+1} + \frac{D}{x-1}$.
 - **Equations and Derivations**
 - **Identity for Equal Degree:** $\frac{2x}{x+1} \equiv A + \frac{B}{x+1}$ (Linear over linear must be a constant).
 - **Identity for Cubic over Quadratic:** $\frac{3x^3}{(x-1)(x-2)} \equiv Ax + B + \frac{C}{x-1} + \frac{D}{x-2}$.
 - **Practical Significance**
 - Crucial step before performing **integration** or **binomial expansion** on top-heavy rational functions.
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **Identity Symbols:**
 - $\equiv$ (Identical to)
- **Linear Factor Form:**
 - $\frac{5}{(x+1)(x-4)} \equiv \frac{A}{x+1} + \frac{B}{x-4}$
- **Three Factor Form:**
 - $\frac{7}{(x-2)(x+6)(x+3)} \equiv \frac{A}{x-2} + \frac{B}{x+6} + \frac{C}{x+3}$
- **Repeated Factor Form:**
 - $\frac{2x+9}{(x-5)(x+3)^2} \equiv \frac{A}{x-5} + \frac{B}{x+3} + \frac{C}{(x+3)^2}$
- **Improper Fraction Forms:**
 - $\frac{x^2}{(x+1)(x-1)} \equiv A + \frac{B}{x+1} + \frac{C}{x-1}$
 - $\frac{x^3}{(x+1)(x-1)} \equiv Ax + B + \frac{C}{x+1} + \frac{D}{x-1}$
- **General Division Relationship:**
 - $F(x) = Q(x) \times \text{divisor} + \text{remainder}$
- **Constants:**

- A, B, C, D, E, F (Constants to be found)
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ULTRA-CONDENSED FINAL SUMMARY

- **Partial Fractions Goal:** Decompose a complex fraction into a sum of simpler linear denominators.
- **Methodology:** Use **substitution** (to eliminate variables) or **equating coefficients** (matching powers) to find unknown constants.
- **Repeated Factors Rule:** A factor $(x + a)^n$ generates n terms:

$$\frac{A_1}{(x + a)} + \frac{A_2}{(x + a)^2} + \dots + \frac{A_n}{(x + a)^n}.$$
- **Improper Fractions Requirement:** If numerator degree $\geq$ denominator degree, you **must** use algebraic division or the $F(x) = Q(x)D(x) + R(x)$ relationship first.
- **High-Yield Check:** Always ensure the final partial fractions match the degree of the quotient (constant for equal degrees, linear for cubic/quadratic).

CHAPTER 3 COORDINATE GEOMETRY IN THE (x, y) PLANE

3.1 Parametric equations

- **Definition / Core Concept**
 - **Parametric Equations:** A system where the x - and y -coordinates of each point on a curve are expressed as functions of a third variable, called a **parameter** (often represented by the letter t).
 - **Cartesian Equation:** An equation involving only the variables x and y in two dimensions.
 - **Domain of the Parameter:** The set of values of t which define the points on the curve.
- **Governing Law / Principle**

- A curve C is defined by $x = p(t)$ and $y = q(t)$.
- Each value of t defines a point with coordinates $(p(t), q(t))$.
- **Conceptual and Mechanistic Explanation (Conversion Mechanism)**
 - **Eliminating the Parameter:** To convert parametric equations to Cartesian form, rearrange one equation to make t the subject (e.g., $t = f(x)$), then substitute this expression into the other equation (e.g., $y = g(t) = g(f(x))$).
 - **Finding Domain and Range:**
 - The **domain** of the Cartesian function $y = f(x)$ is the **range** of the parametric function for x , which is $p(t)$.
 - The **range** of the Cartesian function $y = f(x)$ is the **range** of the parametric function for y , which is $q(t)$.
- **Assumptions, Conditions, and Constraints**
 - **Watch Out (Coordinate Disparity):** The value of the parameter t is generally **not equal** to either the x - or the y -coordinate.
 - **Watch Out (Multi-valued Coordinates):** More than one point on the curve can have the same x -coordinate or same y -coordinate for different values of t .
 - **Watch Out (Domain Limits):** Do not draw any points on the curve that fall outside the range of x and y values defined by the given domain of t .
 - **Inverse Functions:** When substituting, use inverse functions (e.g., e^x is the inverse of $\ln(x + 3)$) to rearrange for t .
- **Special Cases and Boundary Scenarios**
 - **Segments:** Straight lines and line segments are referred to as 'curves' in coordinate geometry.
 - **Empty Domain:** If the domain of t is given as $\mathbb{R}$, it implies t can take any real value.
 - **Endpoints:** When t is restricted (e.g., $-3 < t < 3$), the resulting Cartesian curve is a specific segment with defined endpoints.

- **Equations and Derivations**

- **Quadratic Curve Derivation:**

$$\begin{array}{l} 2. \quad x = 2t \implies t = \frac{x}{2} \end{array}$$

$$\begin{array}{l} 3. \quad y = t^2 \implies y = \left(\frac{x}{2}\right)^2 = \frac{x^2}{4} \end{array}$$

- **Rational Function Derivation:**

$$2. \quad x = \ln(t + 3) \implies e^x = t + 3 \implies t =$$

$$\begin{array}{l} 3. \quad y = \frac{1}{t + 5} \implies y = \frac{1}{(e^x - 3) + 5} = \end{array}$$

- **Diagram Integration**

- [Diagram: Graph of a curve on (x, y) axes showing various points labeled with their corresponding parameter values, e.g., $t = 2, t = 0.5$. Axes labeled 0 to 6 for x and -1 to 7 for y .]

- [Graph: Parabola $y = \frac{x^2}{4}$ restricted between $x = -6$ and $x = 6$, with range $0 \leq y < 9$.]

3.2 Using trigonometric identities

- **Definition / Core Concept**

- **Trigonometric Parametric Equations:** Equations where x and y are defined using circular functions ($\sin, \cos, \tan, \sec, \text{etc.}$).
- **Radians Requirement:** In this chapter, angles are always measured in **radians**.

- **Governing Law / Principle**

- Utilize **Pythagorean identities** and **double-angle formulae** to eliminate the parameter t .

- **Key Identity 1:** $\sin^2 t + \cos^2 t \equiv 1$.

- **Key Identity 2:** $1 + \tan^2 t \equiv \sec^2 t$ (rearranged as $\sec^2 t - \tan^2 t \equiv 1$).

- **Key Identity 3:** $1 + \cot^2 t \equiv \csc^2 t$.

- **Conceptual and Mechanistic Explanation**

- **Elimination Technique:**

- Express the trig functions (e.g., $\sin t$ and $\cos t$) in terms of x and y .

- Substitute these expressions into the appropriate identity.

- Ensure you square the whole expression when substituting into

identities like $\sin^2 t + \cos^2 t \equiv 1$.

- **Assumptions, Conditions, and Constraints**

- **Watch Out (Square Roots):** Be careful when taking square roots (e.g.,

$$\cos t = \sqrt{1 - \sin^2 t}).$$

You must consider the range of the parameter t to decide if the root is positive or negative.

- **Special Cases / Geometric Significance**

- **Circles:** Parametric equations $x = a + r \cos t$, $y = b + r \sin t$ represent a circle with center (a, b) and radius r .

- **Standard Circle Form:** $(x - a)^2 + (y - b)^2 = r^2$.

- **Ellipses:** Derived from forms like

$$x = 3 \cos t + 4, y = 2 \sin t.$$

- **Equations and Derivations**

- **Deriving a Circle Identity:**

- $x = \sin t + 2 \implies \sin t = x - 2$
- $y = \cos t - 3 \implies \cos t = y + 3$
- Substitute into $\sin^2 t + \cos^2 t \equiv 1$.
- Result: $(x - 2)^2 + (y + 3)^2 = 1$.
- **Using Double Angle Identity:**
 - Given $x = \sin t, y = \sin 2t$
 - $y = 2 \sin t \cos t$
 - $\cos t = \sqrt{1 - \sin^2 t} = \sqrt{1 - x^2}$
 - Result: $y = 2x\sqrt{1 - x^2}$.
- **Using Sec/Tan Identity:**
 - $x = \cot t + 2 \implies \cot t = x - 2 \implies \cot^2 t = (x - 2)^2$
 - $y = \csc^2 t - 2 \implies \csc^2 t = y + 2$
 - Substitute into $1 + \cot^2 t \equiv \csc^2 t$:
 - $1 + (x - 2)^2 = y + 2 \implies y = (x - 2)^2 - 1$

- **Diagram Integration**

- [Diagram: Circle on a coordinate grid with center at $(2, -3)$ and radius 1].
- [Graph: Quadratic curve $y = x^2 - 4x + 3$ with minimum point at $(2, -1)$ and y -intercept at 3].

3.3 Curve sketching

- **Definition / Core Concept**

- **Curve Sketching:** Plotting a parametric curve by substituting values of the parameter into each equation to generate a table of coordinates.

- **Governing Law / Principle**

- For a defined domain of t , calculate corresponding values of x and y to find points on the curve.

- **Conceptual and Mechanistic Explanation (Method)**

- Choose values for t covering the entire given domain.
- Create a table: row 1 (t), row 2 ($x(t)$), row 3 ($y(t)$).
- Plot the resulting coordinates (x, y) on the plane.
- Draw a smooth curve through the points.

- **Assumptions, Conditions, and Constraints**

- **Partial Graphs:** If the domain of t is restricted (e.g., $-1 \leq t \leq 5$), the resulting sketch is only a portion of the Cartesian curve.
- **Technology Use:** Graphic calculators or software (e.g., GeoGebra) can be used to visualize complicated parametric curves.

- **Special Cases / Behavior**

- **Direction of Motion:** As t increases or decreases, the curve "moves" in a specific direction.
- **Asymptotic Behavior:** Curves may approach infinity as t approaches specific limits (e.g., as $t \rightarrow -1$).

- **Equations and Examples**

- **Ellipse Sketching:**

$$x = 3 \cos t + 4, y = 2 \sin t, 0 \leq t \leq 2\pi$$

- Result: Coordinates include

$$(7, 0), (4, 2), (1, 0), (4, -2).$$

- **Partial Quadratic Sketching:**

$$x = 2t, y = t^2, -1 \leq t \leq 5$$

- Result: Graph starts at $(-2, 1)$ and ends at $(10, 25)$.

- **Diagram Integration**

- [Diagram: Table of values for an ellipse: t values $0, \frac{\pi}{4}, \frac{\pi}{2} \dots 2\pi$ and corresponding x, y values].
- [Graph: Ellipse centered at $(4, 0)$ with major axis 6 and minor axis 4].
- [Graph: Segment of the parabola $y = \frac{x^2}{4}$ from $x = -2$ to $x = 10$].

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Basic Forms:**

- $x = p(t), y = q(t)$ (Parametric Equations)
- $y = f(x)$ (Cartesian Equation)

- **Trigonometric Identities:**

- $\sin^2 t + \cos^2 t \equiv 1$
- $1 + \tan^2 t \equiv \sec^2 t$
- $1 + \cot^2 t \equiv \csc^2 t$
- $\sin 2t \equiv 2 \sin t \cos t$

- **Geometric Shapes:**

- $(x - a)^2 + (y - b)^2 = r^2$ (Circle with center (a, b) and radius r)
- $\frac{(x - 4)^2}{3^2} + \frac{y^2}{2^2} = 1$ (Example Ellipse)

- **Domain and Range Mapping:**

- Domain of $f(x) = \text{Range of } p(t)$
 - Range of $f(x) = \text{Range of } q(t)$
 - **Sets:**
 - $\mathbb{R}$ (Set of all real numbers)
 - t (Parameter variable)
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ULTRA-CONDENSED FINAL SUMMARY

- **Parametric Basics:** Express x and y as functions of a third variable t . A single value of t generates one coordinate pair (x, y) .
- **Cartesian Conversion:** Rearrange one parametric equation for t and substitute into the other. Use inverse functions where necessary.
- **Domain/Range Rule:** The domain of the Cartesian result is strictly determined by the range of the x -parametric function over the allowed values of t .
- **Trigonometric Elimination:** Use Pythagorean identities ($\sin^2 + \cos^2 = 1$) to remove trig parameters. Ensure angles are in radians.
- **Circle Representation:**
 $x = a + r \cos t, y = b + r \sin t$ is a circle of radius r centered at (a, b) .
- **Sketching Procedure:** Create a table of values for $t, x,$ and y . Plot the resulting coordinates. Be mindful of parameter-driven boundaries.

CHAPTER 4 BINOMIAL EXPANSION

4.1 Expanding $(1 + x)^n$

- **Definition / Core Concept**

- **Binomial Expansion (Infinite):** For fractional or negative values of n , the expansion of $(1 + x)^n$ results in an infinite series rather than a finite number of terms.
- **Convergent Series:** A series where the sum of terms approaches a finite limit as the number of terms increases; required for the expansion to be valid.

- **Governing Law / Principle**

- **The Binomial Series Formula:**

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots + \frac{n(n-1)\dots(n-r+1)}{r!}x^r + \dots$$

- **Conceptual and Mechanistic Explanation (Why it works)**

- **Infinite Nature:** When n is not a natural number (positive integer), none of the factors in the expression $n(n-1)\dots(n-r+1)$ will ever equal zero. Consequently, the coefficients never become zero, and the series continues indefinitely.
- **Approximation Mechanism:** For small values of x , higher powers of x (e.g., x^3, x^4) become increasingly insignificant, allowing the first few terms to provide a highly accurate approximation of the function.

- **Assumptions, Conditions, and Constraints**

- **Validity Condition:** The expansion is valid if and only if $|x| < 1$, where $n \in \mathbb{R}$.
- **Interval of Validity:** This is equivalent to $-1 < x < 1$.
- **Modified Variable:** For the expansion of $(1 + bx)^n$, the series is valid when

$$|bx| < 1, \text{ which rearranges to } |x| < \frac{1}{|b|}.$$

- **Special Cases and Edge Behavior**

- **Natural Numbers:** If n is a natural number ($n \in \mathbb{N}$), the formula

$$(a + b)^n = a^n + \binom{n}{1}a^{n-1}b + \dots + b^n$$

produces a finite number of terms $(n + 1)$.

- **Sign Handling:** When expanding $(1 - x)^n$, replace x with $(-x)$. Even powers of $(-x)$ result in positive terms; odd powers result in negative terms.

- **Equations and Derivations**

$$\frac{n(n-1)\dots(n-r+1)}{r!}x^r$$

- **General Term:**

$$\frac{1}{1+x}$$

- **Index Form Conversion:** Fractions like $\frac{1}{1+x}$ must be written as $(1+x)^{-1}$ before expanding.

- **Surd Approximation:** To approximate $\sqrt{2}$, use the expansion of $\sqrt{1-2x}$ and substitute a small value such as $x = 0.01$ to get

$$\sqrt{0.98} = \frac{\sqrt{98}}{\sqrt{100}} = \frac{7\sqrt{2}}{10}$$

- **Practical Significance**

- **Medical Physics:** Used by physicists to analyze magnetic fields in MRI scanners via polynomial approximations of expressions involving fractional and negative indices.

4.2 Expanding $(a + bx)^n$

- **Definition / Core Concept**

- **General Binomial Form:** Expanding expressions where the first term in the bracket is a constant $a \neq 1$.

- **Governing Law / Principle**

- To expand $(a + bx)^n$ where n is negative or a fraction, you **must** first factor out a^n to transform the expression into the form $(1 + X)^n$.

- **Conceptual and Mechanistic Explanation**

- **Factoring Step:**

$$(a + bx)^n = \left(a \left(1 + \frac{b}{a}x \right) \right)^n = a^n \left(1 + \frac{b}{a}x \right)^n$$

- **Distribution:** Ensure a^n is multiplied by **every term** in the resulting expansion

$$\text{of } \left(1 + \frac{b}{a}x \right)^n$$

- **Assumptions, Conditions, and Constraints**

- **Validity Range:** The expansion converges and is valid only when $\left| \frac{b}{a}x \right| < 1$.
 - **Rearranged Constraint:** This is equivalent to $|x| < \left| \frac{a}{b} \right|$.
- **Equations and Derivations**
 - **Expansion Process (Example $\sqrt{4+x}$):**
 1. Write in index form: $(4+x)^{\frac{1}{2}}$.
 2. Factor out $4^{\frac{1}{2}}$: $4^{\frac{1}{2}}\left(1+\frac{x}{4}\right)^{\frac{1}{2}} = 2\left(1+\frac{x}{4}\right)^{\frac{1}{2}}$.
 3. Expand $\left(1+\frac{x}{4}\right)^{\frac{1}{2}}$ using the binomial series formula.
 4. Multiply final result by 2.
- **Diagram Integration**
 - [Graph: Number line showing the region of convergence for an expansion, e.g., highlighting the interval between $-\frac{2}{3}$ and $\frac{2}{3}$ for $|3x| < 2$].

4.3 Using partial fractions

- **Definition / Core Concept**
 - **Decomposition for Expansion:** Using partial fractions to break down complex algebraic fractions into a sum of simpler terms that can each be expanded binomially.
- **Governing Law / Principle**

$$\frac{f(x)}{g(x)}$$
 - If an expression is of the form $\frac{f(x)}{g(x)}$ where $g(x)$ can be split into linear factors, split it into partial fractions **before** expanding each part separately.
- **Conceptual and Mechanistic Explanation**
 - Step 1: Convert the original fraction into partial fractions.
 - Step 2: Rewrite each partial fraction in index form (e.g., $A(1+x)^{-1} + B(2-x)^{-1}$).
 - Step 3: Factor out constants to get each term in $(1+X)^n$ form.
 - Step 4: Expand each term separately to the required power of x .
 - Step 5: Sum the resulting series expansions.

- **Assumptions, Conditions, and Constraints**

- **Combined Validity:** To find the range of values of x for which the *entire* expansion is valid, you must find the set of values that satisfy **both** (or all) individual inequalities.
- **Intersection of Sets:** If one part is valid for $|x| < 1$ and another for $|x| < 2$, the total expansion is valid only for $|x| < 1$ (the more restrictive condition).

- **Practical Significance**

- Allows for the approximation of highly complex rational functions used in engineering and physics modeling.

CONSOLIDATED EQUATION AND SYMBOL LIST

- **The Binomial Series Formula** ($n \in \mathbb{R}$):

- $$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \frac{n(n-1)(n-2)}{3!}x^3 + \dots + \dots$$

- **The General Term:**

- $$\frac{n(n-1)\dots(n-r+1)}{r!}x^r$$

- **Validity Conditions:**

- $|x| < 1$ (For $(1+x)^n$).
- $|bx| < 1 \implies |x| < \frac{1}{|b|}$ (For $(1+bx)^n$).
- $\left|\frac{b}{a}x\right| < 1 \implies |x| < \left|\frac{a}{b}\right|$ (For $(a+bx)^n$).

- **Transformation Formula:**

- $$(a+bx)^n = a^n \left(1 + \frac{b}{a}x\right)^n$$

- **Partial Fraction Form:**

- $$\frac{f(x)}{g(x)} \equiv \frac{A}{x-a} + \frac{B}{x-b} \dots$$

- **Constants and Sets:**
 - n (Index, can be negative or fractional).
 - $r!$ (Factorial: $r \times (r - 1) \times \dots \times 1$).
 - $\mathbb{R}$ (Set of real numbers).
 - $\binom{n}{r}$ (Binomial coefficient; used only for natural numbers n in this context).
-

ULTRA-CONDENSED FINAL SUMMARY

- **Binomial Series:** Used for expanding $(1 + x)^n$ when n is a fraction or negative; results in an **infinite series**.

$$(1 + x)^n = 1 + nx + \frac{n(n-1)}{2!}x^2 + \dots$$

- **The Formula:**
- **Condition of Validity:** Crucial for convergence; expansion is only valid when the magnitude of the variable term is **less than 1** (e.g., $|x| < 1$).
- **Scaling Rule:** For $(a + bx)^n$, you **must** factor out a^n before applying the series formula.
- **Complex Expressions:** Use **partial fractions** to decompose complex rational functions before expanding them as a sum of binomial series.
- **Global Validity:** The valid range for a sum of expansions is the **intersection** (most restrictive part) of the individual ranges.

CHAPTER 5 DIFFERENTIATION

5.1 Parametric differentiation

- **Definition / Core Concept**

$$\frac{dy}{dx}$$

- **Parametric Gradient:** Finding the gradient $\left(\frac{dy}{dx}\right)$ at a given point on a curve where x and y are defined as functions of a third variable (parameter), t , without first converting to Cartesian form.

- **Governing Law / Principle**

- **Chain Rule Variation:** If x and y are given as functions of a parameter t , then:

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt}.$$

- **Conceptual and Mechanistic Explanation**

- **Step 1:** Differentiate x and y separately with respect to the parameter (t or θ)

$$\frac{dx}{dt} \quad \frac{dy}{dt}$$

to find $\frac{dx}{dt}$ and $\frac{dy}{dt}$.

- **Step 2:** Apply the chain rule in quotient form to find the gradient function in terms of the parameter.
- **Step 3:** Substitute the specific value of the parameter to find the numerical gradient at the point.

- **Assumptions, Conditions, and Constraints**

$$\frac{dx}{dt} \neq 0$$

- The derivative $\frac{dx}{dt}$ must not be zero ($\frac{dx}{dt} \neq 0$) for the gradient to be defined.
- Angles in trigonometric parametric equations are measured in **radians**.

- **Special Cases / Edge Behavior**

- **Normals:** The normal is perpendicular to the tangent. If the tangent gradient is

$$m, \text{ the normal gradient is } -\frac{1}{m}.$$

- **Stationary Points:** Points of zero gradient occur where $\frac{dy}{dx} = 0$. This is

$$\frac{dy}{dt} = 0 \quad \left(\text{provided } \frac{dx}{dt} \neq 0 \right).$$

- **Finding Coordinates:** To find the (x, y) coordinates of a point given a parameter value, substitute the value back into the original parametric equations.

- **Equations and Derivations**

- **Equation of Normal:** $y - y_1 = m(x - x_1)$ where

$$m = -\frac{dx/dt}{dy/dt}.$$

- **Chain Rule Expansion:** $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}.$

- **Practical Significance**
 - Used to model the velocity and displacement of objects, such as a wrecking ball, by differentiating its position functions.

5.2 Implicit differentiation

- **Definition / Core Concept**
 - **Implicit Function:** A function defined by an equation in the form $f(x, y) = g(x, y)$ where y is not isolated as the subject (e.g., $x^2 + 2xy + y^3 = 2$).
 - **Implicit Differentiation:** The process of differentiating such equations term by term with respect to x without rearranging them first.
- **Governing Law / Principle**
 - **General Chain Rule for y :**

$$\frac{d}{dx}(f(y)) = f'(y) \frac{dy}{dx}.$$
- **Conceptual and Mechanistic Explanation**
 - **Mechanism:** When differentiating a term containing y with respect to x , differentiate with respect to y as normal and then multiply by $\frac{dy}{dx}$.
 - **Product Rule Application:** For terms like xy , the product rule must be used:
$$\frac{d}{dx}(xy) = x \frac{dy}{dx} + y(1).$$
 - **Solving for Gradient:** After differentiating all terms, collect all terms involving $\frac{dy}{dx}$ on one side and factorize to make $\frac{dy}{dx}$ the subject.
- **Assumptions, Conditions, and Constraints**
 - **Watch Out:** You must pay careful attention to the variable you are differentiating with respect to.
- **Equations and Specific Results**
 - **Power Rule (Implicit):** $\frac{d}{dx}(y^n) = ny^{n-1} \frac{dy}{dx}.$

- **Example (Exponential/Log):** $\frac{d}{dx}(e^{2x} \ln y)$ requires the product rule:

$$e^{2x} \cdot \frac{1}{y} \frac{dy}{dx} + \ln y \cdot 2e^{2x}.$$
- **Practical Significance**
 - Allows finding the gradient, tangents, and normals for complex geometric curves (like foliums or lemniscates) where explicit form is impossible or difficult to obtain.

5.3 Rates of change

- **Definition / Core Concept**
 - **Connected Rates of Change:** Problems involving multiple variables changing with respect to time, linked by geometric or physical relationships.
 - **Differential Equation:** An equation that involves a rate of change (a derivative).
- **Governing Law / Principle**
 - **Chain Rule Connectivity:**

$$\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}.$$
- **Conceptual and Mechanistic Explanation**
 - **The Chain Rule Link:** The chain rule connects the rate of change of one quantity (e.g., Area A) to another (e.g., radius r) using their derivative relationship.
$$\frac{dr}{dV} = 1 \div \frac{dV}{dr}.$$
 - **Inverse Property:**
- **Assumptions, Conditions, and Constraints**
 - **Direction of Change:** A rate of **decrease** is represented by a **negative** sign (e.g., radioactive decay or cooling).
 - **Proportionality ($\propto$):**
 - Directly proportional to X : $\frac{dy}{dt} = kX$.
 - Inversely proportional to X^2 : $\frac{dy}{dt} = \frac{k}{X^2}$.
- **Special Cases / Modeling Laws**
 - **Radioactive Decay:** $\frac{dN}{dt} = -kN$, where N is the number of particles.

- **Newton's Law of Cooling:** $\frac{d\theta}{dt} = -k(\theta - \theta_0)$, where θ_0 is the surrounding temperature.
 - **Geometric Shapes:**
 - Circle Area: $A = \pi r^2 \implies \frac{dA}{dr} = 2\pi r$.
 - Sphere Volume: $V = \frac{4}{3}\pi r^3 \implies \frac{dV}{dr} = 4\pi r^2$.
 - Hemisphere: $V = \frac{2}{3}\pi r^3$; $S_{total} = 3\pi r^2$.
 - **Practical Significance**
 - Used in medical physics (analyzing MRI magnetic fields), engineering (wrecking ball displacement), and environmental science (ice thickness on ponds or liquid flow in tanks).
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **Parametric Differentiation:**
 - $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$
 - $\frac{dy}{dx} = \frac{dy}{d\theta} \div \frac{dx}{d\theta}$
- **Implicit Differentiation Rules:**
 - $\frac{d}{dx}(f(y)) = f'(y) \frac{dy}{dx}$
 - $\frac{d}{dx}(y^n) = ny^{n-1} \frac{dy}{dx}$
 - $\frac{d}{dx}(xy) = x \frac{dy}{dx} + y$
- **Connected Rates of Change (Chain Rule):**
 - $\frac{dA}{dt} = \frac{dA}{dr} \times \frac{dr}{dt}$
 - $\frac{dr}{dV} = \frac{1}{dV/dr}$
- **Modeling and Proportionality:**
 - $\frac{dN}{dt} = -kN$ (Decay, $k > 0$)
 - $\frac{d\theta}{dt} = -k(\theta - \theta_0)$ (Newton's Cooling)

- $\frac{dx}{dt} = \frac{k}{x^2}$ (Inverse proportionality to square)
 - **Geometric Formulas:**
 - $A = \pi r^2$ (Circle Area)
 - $C = 2\pi r$ (Circle Circumference)
 - $V = \frac{4}{3}\pi r^3$ (Sphere Volume)
 - $S = 4\pi r^2$ (Sphere Surface Area)
 - $V = \frac{1}{3}\pi h^3 \tan^2 \theta$ (Cone Volume - specific context)
 - **Variables and Constants:**
 - t : Time or parameter
 - θ : Angle (radians) or temperature
 - k : Constant of proportionality
 - (x, y) : Coordinates
 - $\frac{dy}{dx}$: Gradient of the tangent
-

ULTRA-CONDENSED FINAL SUMMARY

- **Parametric:** Differentiate $y(t)$ and $x(t)$ separately, then divide $(dy/dt)/(dx/dt)$. Ensure the denominator is non-zero.
- **Implicit:** Use $\frac{d}{dx}(f(y)) = f'(y)\frac{dy}{dx}$ for all y terms. Use product rule for mixed terms like xy . Rearrange for dy/dx .
- **Rates of Change:** Connect rates using the chain rule (e.g., $dy/dt = (dy/dx) \cdot (dx/dt)$).
- **Proportionality in Modeling:** Set up derivatives using k (constant). Always use negative signs for decreasing rates (decay, evaporation, cooling).
- **Normals/Tangents:** Standard line equations $y - y_1 = m(x - x_1)$ apply; for normals, use the negative reciprocal gradient.
- **Radians:** All trigonometric differentiation requires angles in radians.

CHAPTER 6 INTEGRATION

6.1 Finding the area under a curve defined parametrically

- **Definition / Core Concept**
 - **Parametric Area Integration:** Extending the calculation of the area under a curve to cases where x and y are defined by a parameter t .
- **Governing Law / Principle**
 - The Cartesian area formula $A = \int_{x_1}^{x_2} y dx$ is adapted by substituting $y = g(t)$ and $dx = f'(t)dt$.
 - **The Parametric Area Formula:** $A = \int_{t=a}^{t=b} g(t)f'(t)dt$.
- **Conceptual and Mechanistic Explanation**
 - **Mechanism:** To find the area, differentiate the function for x to obtain an expression for the operator dx in terms of dt . Substitute the expressions for y and dx into the integral.
 - **Limits Conversion:** The x -limits x_1 and x_2 must be converted into their corresponding parameter values $t = a$ and $t = b$.
- **Assumptions, Conditions, and Constraints**
 - **Functionality:** $x = f(t)$ and $y = g(t)$ must be continuous over the interval $a \leq t \leq b$.
 - **Direction of Integration:** Ensure the limits of integration (a and b) correspond correctly to the direction of the curve from x_1 to x_2 .
- **Equations and Derivations**
 - **Step 1:** $x = f(t) \implies \frac{dx}{dt} = f'(t) \implies dx = f'(t)dt$.
 - **Step 2:** $A = \int y dx = \int g(t)f'(t)dt$.
 - **Example (Trig):** For $x = \cos t$, $y = -\cos t$, area $= \int (-\cos t)(-\sin t)dt = \int \frac{1}{2} \sin 2t dt$.
- **Practical Significance**
 - Allows for area calculation of complex paths (e.g., orbits, projectile motion) directly from their motion equations.

6.2 Volumes of revolution around the x -axis

- **Definition / Core Concept**

- **Volume of Revolution:** The volume of the 3D solid generated when a region R , bounded by a curve and the x -axis, is rotated through 360° (or 2π radians) about the x -axis.

- **Governing Law / Principle**

$$V = \pi \int_a^b y^2 dx$$

- **Standard Cartesian Formula:**

$$V = \pi \int_{t=a}^{t=b} (g(t))^2 f'(t) dt$$

- **Parametric Volume Formula:**

- **Conceptual and Mechanistic Explanation**

- **The Disc Method:** Consider the solid as an infinite number of small cylindrical strips (discs) of radius y and thickness δx .

- **Volume of Strip:** $\delta V \approx \pi y^2 \delta x$.

- **Integration as Limit:** Total volume is the limit of the sum of these discs as

$$\delta x \rightarrow 0, \text{ yielding } \int \pi y^2 dx$$

- **Assumptions, Conditions, and Constraints**

- **Rotation Range:** The formula specifically assumes a full rotation of 2π radians.
- **Boundaries:** The region must be bounded by a positive curve, the x -axis, and two vertical lines $x = a$ and $x = b$.

- **Special Cases / Behavior**

- **Parametric Transition:** When using parametric forms, the integral is converted purely into terms of the parameter t .

- **Equations and Derivations**

- **General Relationship:**

$$\text{Volume} = \sum \pi y^2 \delta x \implies \pi \int y^2 dx$$

- **Parametric Derivation:** If $x = f(t)$, then $dx = f'(t)dt$. Substituting

$$V = \int \pi (g(t))^2 f'(t) dt$$

gives

- **Practical Significance**

- Used in engineering and manufacturing to calculate the volume and mass of rotational parts (e.g., pistons, bowls, spindles).

6.3 Integration by substitution

- **Definition / Core Concept**
 - **Integration by Substitution:** A technique to simplify an integral by changing the variable, effectively reversing the chain rule of differentiation.
- **Governing Law / Principle**

$$\int f(x)dx = \int f(x)\frac{dx}{du}du$$
 - **The Relationship:**
- **Conceptual and Mechanistic Explanation**
 - **Step 1:** Choose a substitution (e.g., $u = f(x)$).
 - **Step 2:** Differentiate to find the relationship between dx and du (e.g., $\frac{du}{dx} = f'(x) \implies dx = \frac{du}{f'(x)}$).
 - **Step 3:** Replace every x term and the dx operator with the equivalent u terms.
 - **Step 4:** Integrate with respect to u .
 - **Step 5:** If indefinite, substitute back to terms of x .
- **Assumptions, Conditions, and Constraints**
 - **Limits Conversion (Definite Integrals):** The x limits must be replaced by their corresponding u values. You do **not** change back to x after integrating a definite integral if you have changed the limits.
 - **Radians:** For trigonometric substitutions, limits must always be in **radians**.
- **Special Cases / Alternative Forms**
 - **Square Root Substitutions:** Often helpful to use $u^2 = f(x)$ instead of $u = \sqrt{f(x)}$ to simplify the derivative process.
- **Equations and Examples**
 - **Example ($u = 2x + 5$):**

$$\int x\sqrt{2x+5}dx \implies \int \left(\frac{u-5}{2}\right)u^{\frac{1}{2}}\frac{1}{2}du$$
 - **Trig Example:**

$$\int \cos x \sin x (1 + \sin x)^3 dx$$

. Let $u = \sin x + 1$.
- **Practical Significance**
 - Foundational tool for solving complex integrals that appear in physics and engineering models.

6.4 Integration by parts

- **Definition / Core Concept**

- **Integration by Parts:** A technique used to integrate the product of two functions by rearranging the product rule of differentiation.

- **Governing Law / Principle**

- **The Formula:**
$$\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$$

- **Conceptual and Mechanistic Explanation**

- **Selection Logic:** You must choose which function to set as u (which will be

$$\frac{dv}{dx}$$

differentiated) and which to set as $\frac{du}{dx}$ (which will be integrated).

- **Strategy (Logarithms):** For expressions like $x^n \ln x$, always let

$$\frac{1}{x}$$

$u = \ln x$ because it simplifies significantly when differentiated ($\frac{1}{x}$).

- **Repeated Application:** Sometimes integration by parts must be applied twice to

$$\int x^2 e^x dx$$

reduce a power (e.g., $\int x^2 e^x dx$).

- **Special Cases / Edge Behavior**

- **Integrating $\ln x$:** To integrate $\int \ln x dx$, write it as $\int 1 \times \ln x dx$, where $u = \ln x$ and $\frac{dv}{dx} = 1$.

- **Cyclic Integrals:** For integrals like $\int e^x \sin x dx$, applying the method twice returns a multiple of the original integral, which can then be solved algebraically.

- **Equations and Derivations**

- **Derivation from Product Rule:**

- $$\frac{d}{dx}(uv) = u \frac{dv}{dx} + v \frac{du}{dx}$$

- $$\int \frac{d}{dx}(uv) dx = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

$$uv = \int u \frac{dv}{dx} dx + \int v \frac{du}{dx} dx$$

- **Practical Significance**

- Essential for analyzing wave functions and signals where products of trigonometric and exponential functions occur.

6.5 Partial fractions

- **Definition / Core Concept**

- **Integration via Partial Fractions:** Decomposing complex algebraic fractions into simpler sums to enable integration.

- **Governing Law / Principle**

- Integrate each term of the partial fraction decomposition separately using standard rules (primarily logarithmic integration).

- **Conceptual and Mechanistic Explanation**

- **Step 1:** Split the expression into partial fractions (distinct linear, repeated linear, or improper).
- **Step 2:** Integrate each component.

- **Logarithmic Rule:** $\int \frac{1}{ax + b} dx = \frac{1}{a} \ln |ax + b| + c$

- **Assumptions, Conditions, and Constraints**

- **Modulus Requirement:** Always use the **modulus** sign when integrating to find a natural logarithm ($\ln |f(x)|$) to ensure the argument is positive.
- **Improper Fractions:** If the degree of the numerator is $\geq$ the degree of the denominator, you **must** perform algebraic division first.

- **Special Cases / Behavior**

- **Repeated Factors:** Remember that $\int \frac{1}{(x-2)^2} dx$ does **not** involve $\ln$
 $-\frac{1}{x-2} + c$
 ; it integrates using the power rule to

- **Equations and Examples**

- **Standard Form:**

$$\int \left(\frac{A}{x+1} + \frac{B}{x-2} \right) dx = A \ln |x+1| + B \ln |x-2| + c$$

- **Log Laws Integration:** Results are often simplified using

$$\ln a + \ln b = \ln ab \text{ or } \ln a - \ln b = \ln \frac{a}{b}.$$

- **Practical Significance**

- Used to solve integration problems involving complex rational functions in chemical kinetics and population modeling.

6.6 Solving differential equations

- **Definition / Core Concept**

- **Differential Equation (DE):** An equation containing derivatives.
- **General Solution:** A solution representing a **family of curves**, containing an arbitrary constant C .
- **Particular Solution:** A specific solution where the constant C is determined using a given point (**boundary condition**).

- **Governing Law / Principle**

- **Separation of Variables:** To solve a first-order DE in the form

$$\frac{dy}{dx} = f(x)g(y), \text{ you must group all } y \text{ terms with } dy \text{ and all } x \text{ terms with } dx.$$

- **The Separated Form:** $\int \frac{1}{g(y)} dy = \int f(x) dx$.

- **Conceptual and Mechanistic Explanation**

- **Mechanism:** Rearrange the equation so one side is purely in terms of y and the other in terms of x . Integrate both sides.
- **Constant Handling:** Only one constant of integration ($+C$) is required, usually added to the x side.

- **Assumptions, Conditions, and Constraints**

- **Domain Validity:** Solutions are valid only where the functions are defined (e.g., $y > 0$ for $\ln y$).

- **Equations and Derivations**

- **Example (General):**

$$\frac{dy}{dx} = 12x^2 - 1 \implies y = 4x^3 - x + C.$$

- **Example (Separated):**

$$\frac{dy}{dx} = x \tan y \implies \int \cot y dy = \int x dx \implies \ln |\sin y| = \frac{1}{2}x^2 + c$$

- **Boundary Condition:** If $x = 1, y = 4$ is given for

$$y - 2 = k \frac{x + 2}{2x + 1}, \text{ substitute to find } k = 2.$$

- **Diagram Integration**

- [Graph: A family of curves (e.g., quadratics) shifted vertically, labeled with different C values like $C = 1, C = 0, C = -1$].

- **Practical Significance**

- Used to find the specific path or state of a system (e.g., an object's trajectory) when its starting point is known.

6.7 Modelling with differential equations

- **Definition / Core Concept**

- **Mathematical Modelling:** Using DEs to represent and predict real-life situations involving change.

- **Governing Law / Principle**

- Translate written descriptions of rates of change into DEs, solve them, and interpret the results in context.

- **Conceptual and Mechanistic Explanation (Formulation)**

- **Growth/Increase:** Rate is proportional to amount ($\frac{dP}{dt} = kP$).

- **Decay/Decrease:** Rate is negative ($\frac{dV}{dt} = -kA$).

- **Connected Rates:** Use the chain rule (e.g., $\frac{dh}{dt} = \frac{dh}{dV} \times \frac{dV}{dt}$).

- **Assumptions, Conditions, and Constraints**

- **Limitations:** Models are approximations. For example, population models ($P = Ae^{kt}$) may not account for space or food limits at large values of t .
- **Units:** Ensure consistency in time (seconds, hours, years) and quantity (mass, volume, count).

- **Special Cases / Examples**

- **Radioactive Decay:** $\frac{dN}{dt} = -kN$.
 - **Newton's Cooling:** $\frac{d\theta}{dt} = -k(\theta - \theta_0)$.
 - **Cylindrical Tank Flow:** Volume $V = \pi r^2 h$. If diameter is 20,
 $V = 100\pi h$, so $\frac{dV}{dh} = 100\pi$.
 - **Equations and Specific Models**
 - **Microorganism Growth:** $P = 8e^{3t}$ (where $P = 8$ at $t = 0$ and $\frac{dP}{dt} = 3P$).
 - **Carbon Dating:** $R = R_0 e^{-kt}$.
 - **Practical Significance**
 - Used by archaeologists to estimate the age of fossils and by scientists to predict population trends or chemical concentrations.
-

CONSOLIDATED EQUATION AND SYMBOL LIST

- **Area and Volume Formulas:**
 - $A = \int_{t=a}^{t=b} y \frac{dx}{dt} dt = \int_{t=a}^{t=b} g(t) f'(t) dt$ (Parametric Area)
 - $V = \pi \int_a^b y^2 dx$ (Cartesian Volume)
 - $V = \pi \int_{t=a}^{t=b} y^2 \frac{dx}{dt} dt$ (Parametric Volume)
- **Integration Techniques:**
 - $\int u \frac{dv}{dx} dx = uv - \int v \frac{du}{dx} dx$ (Integration by Parts)

- $\int f(g(x))g'(x)dx = \int f(u)du$ (Substitution)
 - $\int \frac{1}{ax+b}dx = \frac{1}{a} \ln |ax+b| + c$ (Partial Fraction Integration)
 - **Differential Equations:**
 - $\int \frac{1}{g(y)}dy = \int f(x)dx$ (Separation of Variables)
 - $\frac{dy}{dt} = ky$ (Direct Proportionality/Growth)
 - $\frac{dy}{dt} = -ky$ (Exponential Decay)
 - **Constants and Symbols:**
 - π : Pi (≈ 3.14159)
 - e : Base of natural logs (≈ 2.71828)
 - t : Time or parameter
 - k : Constant of proportionality
 - C : Constant of integration
 - a, b : Limits of integration
-

ULTRA-CONDENSED FINAL SUMMARY

- **Parametric Calculus:** Area is $\int y \frac{dx}{dt} dt$ and Volume is $\pi \int y^2 \frac{dx}{dt} dt$.
Always convert limits to parameter values.
- **Substitution:** Replace every x and dx with u equivalents. Change limits for definite integrals.
- $uv - \int vdu$
- **Parts:** Use for products. Prioritize $u = \ln x$; otherwise, pick u to be the power of x .
- **Rational Functions:** Use partial fractions first. Watch for improper fractions (divide first) and modulus signs in $\ln$ results.

- **DEs:** Solve by separating variables ($\int f(y)dy = \int g(x)dx$). Use boundary conditions to find the particular solution constant C .
- **Modelling:** Translate "rate of change proportional to..." into d/dt equations. Growth is $+k$, decay/cooling is $-k$. Check for model limitations at large t .

CHAPTER 7 VECTORS

7.1 Vectors

- **Definition / Core Concept**
 - **Vector:** A quantity that possesses both magnitude (size) and direction.
 - **Scalar:** A quantity that has magnitude but no direction (e.g., real numbers).
 - **Directed Line Segment:** A representation of a vector from an initial point P to a terminal point Q , denoted as $\vec{PQ}$.
 - **Notation:** Vectors are written as bold lowercase letters (e.g., **a**) in print or underlined (e.g., a) in handwriting.
- **Governing Laws and Principles**
 - **Equality:** Two vectors are equal if they have the same magnitude and direction, even if they start at different points.
 - **Triangle Law for Addition:** The sum of two vectors $\vec{AB}$ and $\vec{BC}$ is the resultant vector $\vec{AC}$, written as $\vec{AB} + \vec{BC} = \vec{AC}$.
 - **Parallelogram Law:** In a parallelogram $ABCD$, where $\vec{AB} = \mathbf{a}$ and $\vec{AD} = \mathbf{b}$, the diagonal $\vec{AC} = \mathbf{a} + \mathbf{b}$.
- **Conceptual and Mechanistic Explanation**
 - **Addition Mechanism:** Adding vectors follows the path from start to finish; $\mathbf{a} + \mathbf{b} = \mathbf{c}$ implies traveling along **a** then **b** results in the same displacement as **c**.

- **Negative Vectors:** The vector $\vec{BA}$ is the negative of $\vec{AB}$, denoted $-\mathbf{a}$, having equal magnitude but opposite direction.
- **Subtraction:** Subtracting is equivalent to adding a negative vector:

$$\mathbf{a} - \mathbf{b} = \mathbf{a} + (-\mathbf{b}).$$
- **Scalar Multiplication:** Multiplying vector $\mathbf{a}$ by scalar k results in a vector with magnitude $|k|$ times the original; it has the same direction if $k > 0$ and opposite if $k < 0$.
- **Special Cases and Edge Behavior**
 - **Zero Vector:** Denoted as $\mathbf{0}$, it occurs when displacement is zero (e.g., $\vec{PQ} + \vec{QP} = \mathbf{0}$).
 - **Parallelism:** Any vector parallel to $\mathbf{a}$ can be written as $\lambda\mathbf{a}$, where λ is a non-zero scalar.
- **Diagram Integration**
 - [Diagram: Directed line segment from P to Q showing vector arrow and label $\mathbf{a}$].
 - [Diagram: Triangle with sides $\mathbf{a}$ and $\mathbf{b}$ showing resultant $\mathbf{c} = \mathbf{a} + \mathbf{b}$].

7.2 Representing vectors

- **Definition / Core Concept**
 - **Displacement:** The change in position relative to axes.
 - **Column Vector:** A vector described by its components, written as $\begin{pmatrix} x \\ y \end{pmatrix}$ in 2D.
 - **Unit Vector:** A vector of length 1.
- **Governing Principles**
 - **Basis Vectors:** In 2D, the unit vectors along the x - and y -axes are

$$\mathbf{i} = \begin{pmatrix} 1 \\ 0 \end{pmatrix} \text{ and } \mathbf{j} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$

- **Cartesian Form:** Any 2D vector can be written as $p\mathbf{i} + q\mathbf{j}$ or as a column

$$\text{vector} \begin{pmatrix} p \\ q \end{pmatrix}.$$

- **Equations and Arithmetic**

- **Scalar Multiplication:** $\lambda \begin{pmatrix} p \\ q \end{pmatrix} = \begin{pmatrix} \lambda p \\ \lambda q \end{pmatrix}.$

- **Addition:** $\begin{pmatrix} p \\ q \end{pmatrix} + \begin{pmatrix} r \\ s \end{pmatrix} = \begin{pmatrix} p + r \\ q + s \end{pmatrix}.$

- **Diagram Integration**

- [Graph: Vector $\mathbf{a}$ on a grid showing a horizontal shift of 3 units ($3\mathbf{i}$) and vertical shift of 4 units ($4\mathbf{j}$)].

7.3 Magnitude and direction

- **Definition / Core Concept**

- **Magnitude:** The length of the vector, denoted $|\mathbf{a}|$.
- **Magnitude-Direction Form:** Defining a vector by its size and the angle it makes with a coordinate axis.

- **Governing Laws / Equations**

$$\mathbf{a} = x\mathbf{i} + y\mathbf{j} = \begin{pmatrix} x \\ y \end{pmatrix}, \text{ magnitude is}$$

- **Pythagoras' Theorem:** For vector

$$|\mathbf{a}| = \sqrt{x^2 + y^2}.$$

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}.$$

- **Unit Vector Formula:** A unit vector in the direction of $\mathbf{a}$ is

$$\tan \theta = \frac{y}{x}$$

- **Direction Angle:** If θ is the angle with the positive x -axis, then

- **Conceptual Explanation**

- **Component Resolution:** A vector with magnitude r and angle α with the $\mathbf{i}$ direction has components $x = r \cos \alpha$ and $y = r \sin \alpha$.
- **Special Cases**
 - If $|\mathbf{a}| = 1$, the vector is already a unit vector.

7.4 Vectors in 3D

- **Definition / Core Concept**
 - **3D Space:** Describes position relative to x , y , and z axes.
 - **3D Unit Vectors:** $\mathbf{i} = (1 \ 0 \ 0)$, $\mathbf{j} = (0 \ 1 \ 0)$, $\mathbf{k} = (0 \ 0 \ 1)$.
- **Governing Principles / Equations**
 - **General Form:** $p\mathbf{i} + q\mathbf{j} + r\mathbf{k} = (p \ q \ r)$.
 - **Magnitude in 3D:** For $\mathbf{a} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$,
 $|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$.
 - **Direction Cosines:** The angles $\theta_x, \theta_y, \theta_z$ with the positive axes satisfy
 $\cos \theta_x = \frac{x}{|\mathbf{a}|}$, $\cos \theta_y = \frac{y}{|\mathbf{a}|}$, and $\cos \theta_z = \frac{z}{|\mathbf{a}|}$.
- **Assumptions and Constraints**
 - 3D vectors obey the same addition and scalar multiplication rules as 2D vectors.

7.5 Solving geometric problems in two dimensions

- **Definition / Core Concept**
 - **Ratio Theorem:** Finding a position vector for a point that divides a line segment in a specific ratio.
- **Governing Principles**

- **Section Formula:** If point P divides AB in ratio $\lambda : \mu$, then

$$\vec{OP} = \vec{OA} + \frac{\lambda}{\lambda + \mu} \vec{AB} = \vec{OA} + \frac{\lambda}{\lambda + \mu} (\vec{OB} - \vec{OA})$$

- **Comparison of Coefficients:** If $\mathbf{a}$ and $\mathbf{b}$ are non-parallel vectors and $p\mathbf{a} + q\mathbf{b} = r\mathbf{a} + s\mathbf{b}$, then $p = r$ and $q = s$.

- **Practical Significance**

- Used to prove geometric properties, such as diagonals of a parallelogram bisecting each other.

7.6 Solving geometric problems in three dimensions

- **Governing Laws**

- **Parallelism:** Vectors in 3D are parallel if one is a scalar multiple of another ($\mathbf{a} = k\mathbf{b}$).
- **Coplanar Vectors:** Vectors that lie in the same plane.
- **Non-coplanar Comparison:** If $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are non-coplanar and $p\mathbf{i} + q\mathbf{j} + r\mathbf{k} = u\mathbf{i} + v\mathbf{j} + w\mathbf{k}$, then $p = u, q = v, r = w$.

- **Special Cases**

- **Trapezium:** A quadrilateral with at least two parallel sides of unequal length.
- **Cuboid/Parallelepiped:** Used as frameworks for 3D vector proofs.

7.7 Position vectors

- **Definition / Core Concept**

- **Position Vector:** A vector giving the position of a point relative to a fixed origin O .
- **Notation:** The position vector of point A is $\vec{OA}$.

- **Governing Principle**

- **Relative Displacement:** For any two points A and B , the vector $\vec{AB}$ is given by $\vec{AB} = \vec{OB} - \vec{OA}$.

- **Conceptual Explanation**

- Traveling from A to B is equivalent to traveling from A to the origin then origin to B ($\vec{AO} + \vec{OB}$).

7.8 3D coordinates

- **Definition / Core Concept**

- **3D Cartesian Coordinates:** Written as (x, y, z) , representing displacement along three mutually perpendicular axes.

- **Governing Equations**

- **Distance from Origin:** $\sqrt{x^2 + y^2 + z^2}$.
- **Distance between Two Points:** Between (x_1, y_1, z_1) and (x_2, y_2, z_2) , distance is $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$.

7.9 Equation of a line in three dimensions

- **Definition / Core Concept**

- **Direction Vector:** A vector $\mathbf{b}$ that is parallel to the line.
- **Scalar Parameter:** Often denoted λ or t , representing the variable that generates different points on the line.

- **Governing Law / Principle**

- **Vector Equation of a Line:** A line passing through point A with position vector $\mathbf{a}$ and parallel to direction vector $\mathbf{b}$ is:

$$\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}.$$

- **Conceptual and Mechanistic Explanation**

- **Mechanism:** Every point on the line is reached by starting at the origin, going to point A , and then moving some multiple λ of the direction vector $\mathbf{b}$.
- **Line through Two Points:** If the line passes through C and D , the direction vector is $\vec{CD} = \mathbf{d} - \mathbf{c}$. The equation is $\mathbf{r} = \mathbf{c} + \lambda(\mathbf{d} - \mathbf{c})$.

- **Rearranged forms**

- The equation can be shown as separate x, y, z components in terms of λ .

7.10 Points of intersection

- **Governing Principle**

- Two lines intersect if there exist parameter values (e.g., λ and μ) that produce the same position vector $\mathbf{r}$ for both lines.

- **Mechanistic Reasoning (The Process)**

- Write the equations of the two lines.
- Equate components to form three linear equations involving the parameters.
- Solve two of the equations simultaneously to find the parameters.
- Substitute these values into the third equation to check for consistency.
- **Result:** If consistent, they intersect. If inconsistent, they do not.

- **Special Cases**

- **Skew Lines:** Lines in 3D that are not parallel and do not intersect.

- **Diagram Integration**

- [Flowchart: Start -> Equate components -> Solve simultaneous equations -> Check 3rd equation -> Consistent? (Yes=Intersect, No=Skew)].

7.11 Scalar product

- **Definition / Core Concept**

- **Scalar Product (Dot Product):** A scalar value derived from two vectors, used to find the angle between them.
- **Notation:** Written as $\mathbf{a} \cdot \mathbf{b}$, said as 'a dot b'.

- **Governing Laws and Equations**

- **Geometric Definition:** $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}| |\mathbf{b}| \cos \theta$, where θ is the angle between the vectors.
- **Component Form:** For $\mathbf{a} = (a_1 \ a_2 \ a_3)$ and $\mathbf{b} = (b_1 \ b_2 \ b_3)$,
 $\mathbf{a} \cdot \mathbf{b} = a_1 b_1 + a_2 b_2 + a_3 b_3$.
- **Perpendicularity Condition:** $\mathbf{a} \cdot \mathbf{b} = 0$ if and only if $\mathbf{a}$ and $\mathbf{b}$ are non-zero and perpendicular ($\cos 90^\circ = 0$).

- **Parallelism Condition:** $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}|$ if they are in the same direction ($\cos 0^\circ = 1$).
 - **Conceptual Explanation**
 - **Angle between lines:** Defined as the acute angle θ between their direction vectors, calculated using

$$\cos \theta = \left| \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|} \right|.$$
 - **Magnitude Relationship:** $\mathbf{a} \cdot \mathbf{a} = |\mathbf{a}|^2$.
 - **Assumptions and Constraints**
 - When finding the angle between two vectors, they must both be directed away from the vertex point.
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CONSOLIDATED EQUATION AND SYMBOL LIST

- **Standard Vector Operations:**
 - $\vec{AB} + \vec{BC} = \vec{AC}$ (Triangle Law)
 - $\vec{AB} = -\vec{BA}$ (Negative Vector)
 - $\vec{AB} = \vec{OB} - \vec{OA}$ (Displacement from Position Vectors)
- **Magnitude and Direction (2D and 3D):**
 - $|\mathbf{a}| = \sqrt{x^2 + y^2}$ (2D Magnitude)
 - $|\mathbf{a}| = \sqrt{x^2 + y^2 + z^2}$ (3D Magnitude)
 - $\hat{\mathbf{a}} = \frac{\mathbf{a}}{|\mathbf{a}|}$ (Unit Vector)
 - $\cos \theta_x = \frac{x}{|\mathbf{a}|}$ (Direction Cosine)
 - Distance = $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$
- **Vector Geometry (Ratios and Lines):**
 - $\vec{OP} = \vec{OA} + \frac{\lambda}{\lambda + \mu} \vec{AB}$ (Section Formula)

- $\mathbf{r} = \mathbf{a} + \lambda \mathbf{b}$ (Vector Equation of a Line)
 - $\mathbf{r} = \mathbf{c} + \lambda(\mathbf{d} - \mathbf{c})$ (Line through two points)
 - **Scalar Product:**
 - $\mathbf{a} \cdot \mathbf{b} = |\mathbf{a}||\mathbf{b}| \cos \theta$ (Geometric Form)
 - $\mathbf{a} \cdot \mathbf{b} = a_1b_1 + a_2b_2 + a_3b_3$ (Cartesian Form)
 - $\cos \theta = \frac{\mathbf{a} \cdot \mathbf{b}}{|\mathbf{a}||\mathbf{b}|}$ (Angle between vectors)
 - **Special Conditions:**
 - $\mathbf{a} \cdot \mathbf{b} = 0 \iff \mathbf{a} \perp \mathbf{b}$ (Perpendicularity)
 - $\mathbf{a} = k\mathbf{b} \iff \mathbf{a} \parallel \mathbf{b}$ (Parallelism)
 - **Symbols:**
 - $\mathbf{i}, \mathbf{j}, \mathbf{k}$: Unit basis vectors
 - λ, μ, t : Scalar parameters
 - $\mathbf{0}$: Zero vector
-

ULTRA-CONDENSED FINAL SUMMARY

- **Vector Basics:** Defined by magnitude and direction. Add via triangle law. Subtract by adding negative. Scalar multiplication changes length.
- **Components:** Represented as $\begin{pmatrix} x \\ y \end{pmatrix}$ or $x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$. Arithmetic is done component-wise.
- **Magnitude/Direction:** Magnitude is via Pythagoras. Unit vectors $(\mathbf{a}/|\mathbf{a}|)$ have length 1. Direction is measured by angles with axes.
- **Lines in 3D:** Defined as $\mathbf{r} = \text{point} + \lambda(\text{direction})$. Two lines intersect if their parameters are consistent across all three component equations.
- **Dot Product:** $\mathbf{a} \cdot \mathbf{b}$ is a scalar. Use it to find angles ($\cos \theta$). Key result: dot product of perpendicular vectors is always **zero**.
- **Geometry:** Position vectors are relative to origin O . Vector $\vec{AB} = \mathbf{b} - \mathbf{a}$. Use non-parallel vector comparison ($p\mathbf{a} + q\mathbf{b} = r\mathbf{a} + s\mathbf{b} \implies p = r, q = s$) to solve proofs.

